Pneumonia Detection

A Dual Machine Learning and Deep Learning Approach

From Chest X-Rays

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github.com/pedromedinatech/pneumonia-detection

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Contents

1 Introduction

Pneumonia is a respiratory infection that affects the lungs and remains a significant cause of mortality, particularly among young children and elderly patients. Chest X-ray imaging is the standard diagnostic tool, but its interpretation depends on radiologist availability and is prone to human error.

This project explores the automatic detection of pneumonia from chest X-ray images by implementing and comparing two independent approaches: a classical Machine Learning pipeline based on handcrafted texture descriptors, and a Deep Learning pipeline based on a pre-trained convolutional neural network. The goal is not to replace clinical diagnosis, but to study how both methodologies handle the same problem and what each one can and cannot capture.

The dataset used is the Chest X-Ray Images (Pneumonia) dataset [1], published by Kermany et al. and available on Kaggle [2]. It contains 5,856 chest X-ray images divided into training, validation, and test splits, each organized into two classes: NORMAL and PNEUMONIA.

1.1 Dataset Analysis

A preliminary exploratory analysis of the dataset revealed three key characteristics that directly influenced the design decisions of both pipelines.

The first finding concerns **class imbalance**. The training set contains 1,342 NORMAL images and 3,876 PNEUMONIA images, representing a ratio of approximately 1:3 in favour of the positive class. This imbalance has a direct impact on classifier behaviour, as models trained on unbalanced datasets tend to develop a bias towards the majority class, producing high recall for PNEUMONIA at the expense of NORMAL recall. Addressing this imbalance was a central concern throughout the project.

The second finding concerns **image dimensionality**. Image resolutions vary significantly across the dataset, ranging from 127 to 2,663 pixels in height and from 384 to 2,916 pixels in width, with average dimensions of approximately 968 x 1,320 pixels. This heterogeneity makes it necessary to apply a uniform resizing step before any feature extraction or model training.

The third finding concerns the **validation split**. The original validation set contains only 8 images per class, totalling 16 images, which is insufficient for meaningful evaluation. To address this, the validation set was merged with the training set throughout the project, and the test set was used exclusively for final evaluation.

Figure 1 illustrates representative examples from both classes. Normal chest X-rays show well-defined lung fields with dark, air-filled regions and clear anatomical structures. Pneumonia cases, by contrast, exhibit characteristic opacities, areas of increased density caused by fluid or inflammatory tissue, particularly in the lower lung zones.

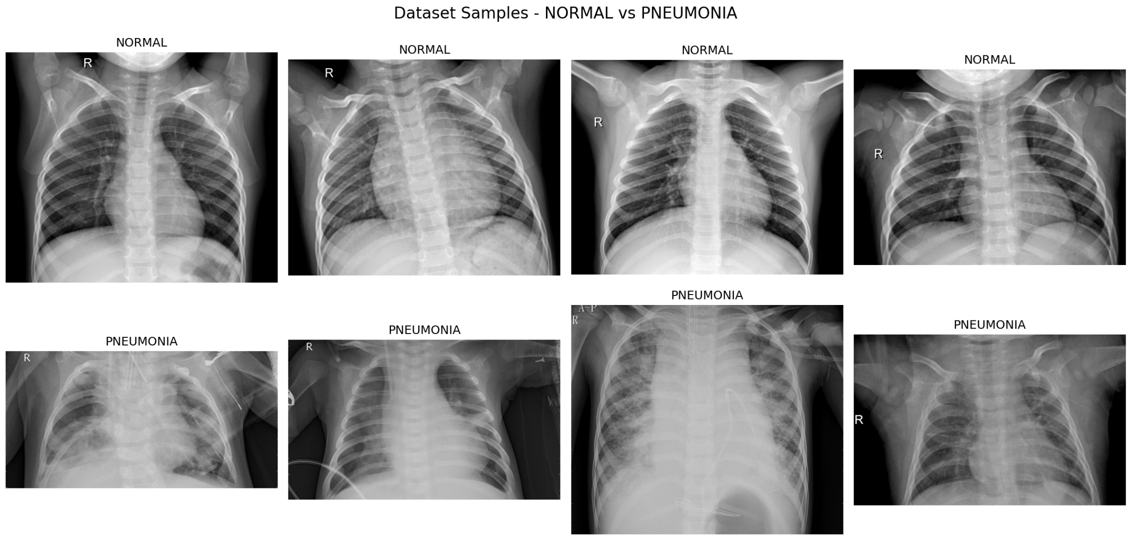


Figure 1: Representative chest X-ray samples from the dataset.

2 Theoretical Background

2.1 Machine Learning for Image Classification

Classical Machine Learning approaches to image classification rely on **feature engineering**: the process of manually designing numerical descriptors that capture relevant information from raw data. Rather than feeding pixel values directly into a classifier, which would result in extremely high-dimensional and noisy input vectors, features are extracted that summarize specific properties of the image. In the context of medical imaging, texture descriptors are particularly useful, as they capture structural patterns in tissue that correlate with pathological conditions.

Once features are extracted, a classifier is trained to learn the mapping between the feature vectors and the corresponding class labels. Two ensemble-based classifiers are used in this project: Random Forest and XGBoost.

2.2 Local Binary Patterns

Local Binary Patterns (LBP) is a texture descriptor that encodes the local structure of an image by comparing each pixel to its surrounding neighbours. For a given pixel, each neighbour is assigned a value of 1 if its intensity is greater than or equal to the central pixel, and 0 otherwise. The resulting binary sequence is converted into a decimal number, producing a single code per pixel.

Formally, for a pixel at position (x, y) with intensity g_c, and P neighbours at radius R, the LBP code is defined as:

$$LBP_{P,R}(x,y) = \sum_{p=0}^{P-1} s(g_p - g_c) \cdot 2^p, \quad s(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

After computing the LBP code for every pixel in the image, a histogram of the resulting codes is built. This histogram serves as the feature vector for the image, capturing the distribution of local micro-patterns such as edges, flat regions, and corners. In this project, LBP is computed with radius $R = 3$ and $P = 24$ neighbours, producing a 26-dimensional histogram.

2.3 Gray-Level Co-occurrence Matrix

The Gray-Level Co-occurrence Matrix (GLCM) is a statistical texture descriptor that captures spatial relationships between pixel intensities. Given an image, the GLCM is a square matrix where each entry (i, j) represents how frequently a pixel with intensity i appears adjacent to a pixel with intensity j, at a given distance and direction.

From the GLCM, four scalar properties are extracted that summarize different aspects of the texture:

Contrast measures the intensity variation between neighbouring pixels. High contrast values indicate large local intensity differences, which in chest X-rays correspond to the presence of opacities adjacent to air-filled regions.

Correlation measures the linear dependency between neighbouring pixel intensities. Uniform tissue produces high correlation, while heterogeneous or inflamed tissue produces lower values.

Energy measures the uniformity of the intensity distribution. Healthy lung tissue, which is predominantly air, tends to produce more uniform distributions and therefore higher energy values.

Homogeneity measures the closeness of the distribution of elements in the GLCM to its diagonal. It is sensitive to small local variations in intensity.

In this project, the GLCM is computed at two distances (1 and 3 pixels) and four angles (0, 45, 90 and 135 degrees), producing 8 values per property and 32 values in total across the four properties.

2.4 Random Forest

Random Forest is an ensemble classifier that builds a large number of decision trees during training, each trained on a random subset of the data and a random subset of the features. The final prediction is determined by majority vote across all trees. This randomness reduces overfitting and improves generalisation compared to a single decision tree.

In this project, Random Forest is trained with 200 trees and `class_weight="balanced"`, which automatically adjusts the weight of each class inversely proportional to its frequency, partially compensating for the class imbalance in the dataset.

2.5 XGBoost

XGBoost (Extreme Gradient Boosting) is an ensemble method based on gradient boosting, where trees are built sequentially, each one correcting the errors of the previous one. Unlike Random Forest, which builds trees in parallel independently, XGBoost optimises a differentiable loss function in a stage-wise manner, producing a strong classifier from a sequence of weak ones.

Class imbalance is handled through the `scale_pos_weight` parameter, which assigns a higher weight to the minority class during training.

2.6 Class Imbalance and SMOTE

The training dataset contains approximately three times more PNEUMONIA images than NORMAL images. This imbalance causes classifiers to develop a bias towards the majority class, producing high recall for PNEUMONIA but poor recall for NORMAL, which is the more clinically dangerous error to miss.

Two strategies are used in this project to address this problem. In the ML pipeline, SMOTE (Synthetic Minority Oversampling Technique) [3] is applied to the feature vectors. SMOTE generates synthetic samples of the minority class by interpolating between existing samples and their nearest neighbours in the feature space, rather than simply duplicating existing ones. This introduces new, unseen points that help the classifier learn a more robust decision boundary for the NORMAL class.

In the DL pipeline, a `WeightedRandomSampler` is used during training to ensure that each batch contains an approximately equal number of samples from both classes, without modifying the dataset itself.

2.7 Convolutional Neural Networks

A Convolutional Neural Network (CNN) is a type of neural network designed to process grid-structured data such as images. Unlike fully connected networks, which would treat each pixel as an independent input, CNNs apply learned filters that slide across the image, detecting local patterns such as edges, textures and shapes. Early layers detect simple low-level features, while deeper layers combine them into increasingly abstract representations.

This hierarchical feature learning makes CNNs particularly well suited for image classification tasks, as they can automatically discover the visual patterns most relevant to the problem without manual feature engineering [7].

2.8 Transfer Learning

Training a deep CNN from scratch requires large amounts of labelled data and significant computational resources. Transfer Learning addresses this by reusing a network that has already been trained on a large dataset, typically ImageNet, which contains over 1.2 million images across 1,000 categories.

The pre-trained network has already learned to detect a wide range of visual features, from low-level edges and textures to high-level shapes and structures. By replacing only the final classification layer and fine-tuning the network on the target dataset, it is possible to achieve strong performance even with relatively few training samples.

In this project, training is carried out in two phases. In the first phase, all layers of the pre-trained network are frozen and only the newly added classification layers are trained. This allows the classifier to adapt to the new task while preserving the pre-trained representations. In the second phase, the last convolutional block is unfrozen and trained with a very small learning rate, allowing the network to make subtle adjustments to its visual representations for the specific domain of chest X-rays.

2.9 DenseNet121

DenseNet121 is a convolutional neural network architecture in which each layer receives the feature maps from all preceding layers as input, rather than only from the immediately previous layer. This dense connectivity pattern encourages feature reuse throughout the network, improves gradient flow during training, and reduces the number of parameters compared to architectures of similar depth [4].

These properties make DenseNet121 particularly well suited for medical image classification, where datasets are often small and subtle visual differences between classes require the preservation of fine-grained features across all network depths. It is worth noting that DenseNet121 was the architecture used in CheXNet, a model developed at Stanford University that demonstrated radiologist-level performance on pneumonia detection from chest X-rays [5].

2.10 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is a technique for visualising which regions of an input image most influenced a neural network's prediction. It produces a heatmap that highlights the areas the network focused on when making its decision, providing a degree of interpretability to otherwise opaque deep learning models [6].

The method works in five steps. First, a forward pass is performed and the predicted class score is obtained. Second, the gradients of the predicted class score with respect to the feature maps of the target convolutional layer are computed. Third, these gradients are globally averaged to obtain a weight for each feature map, representing its importance to the prediction. Fourth, a weighted combination of the feature maps is computed and passed through a ReLU activation, retaining only the regions that contributed positively to the predicted class. Fifth, the resulting low-resolution map is upsampled to the original image size and overlaid as a colour heatmap.

In this project, Grad-CAM is applied to the last convolutional layer of DenseNet121, `denseblock4.denselayer16.conv2`, which contains the most semantically rich representations.

3 Methodology

3.1 ML Pipeline: Feature Extraction and Classification

The Machine Learning pipeline treats each chest X-ray as a source of statistical texture information rather than a raw pixel array. Every image is first converted to grayscale and resized to 224 x 224 pixels to ensure uniform dimensionality across the dataset. Two texture descriptors are then computed and concatenated into a single feature vector per image.

LBP is computed with radius $R = 3$ and $P = 24$ neighbours using the uniform mapping, which reduces the number of distinct patterns and improves robustness to noise. The resulting histogram contains 26 values. The GLCM is computed at distances 1 and 3 pixels and at angles 0, 45, 90 and 135 degrees. Four properties are extracted from each GLCM: contrast, correlation, energy and homogeneity, producing 32 values in total. The final feature vector has 58 dimensions.

Two classifiers are trained on these vectors: Random Forest with 200 estimators and XGBoost with 200 estimators. Both are evaluated on the held-out test set. To address class imbalance, SMOTE is applied to the training set after feature extraction, generating synthetic NORMAL samples until both classes are equally represented. The effect of SMOTE is measured by comparing results before and after its application.

3.2 DL Pipeline: Two-Phase Training Strategy

The Deep Learning pipeline uses DenseNet121 pre-trained on ImageNet as a feature extractor. The original classification head, designed for 1,000 classes, is replaced by a custom sequential block consisting of a fully connected layer with 256 units and ReLU activation, a dropout layer with rate 0.3, and a final output layer with 2 units and softmax activation corresponding to the NORMAL and PNEUMONIA classes.

Training is carried out in two phases. In Phase 1, all layers of the DenseNet121 backbone are frozen and only the custom classification head is trained for 15 epochs with a learning rate of 10^{-3} and the Adam optimiser. This allows the classifier to adapt to the binary task while the pre-trained representations remain intact.

In Phase 2, the last convolutional block of DenseNet121 (`denseblock4`) and the final batch normalisation layer (`norm5`) are unfrozen. Training continues for 10 additional epochs with a significantly reduced learning rate of 10^{-5}, allowing the network to make subtle domain-specific adjustments to its representations without overwriting the knowledge acquired during ImageNet pre-training. Table 1 summarises the training configuration for both phases.

Table 1: Training configuration for each phase.

Parameter	Phase 1	Phase 2
Epochs	15	10
Learning rate	10^{-3}	10^{-5}
Optimiser	Adam	Adam
Batch size	32	32
Frozen layers	All backbone	All except denseblock4 + norm5
Trainable parameters	262,914	2,423,042

Data augmentation is applied during training through random horizontal flips, rotations of up to 10 degrees, and small variations in brightness and contrast. These transformations increase the effective diversity of the training set and reduce overfitting without modifying the original images on disk. Class imbalance is handled via a `WeightedRandomSampler`, which adjusts the sampling probability of each image so that both classes are represented equally within each training batch.

3.3 Addressing Class Imbalance: Strategies and Findings

Three distinct strategies were explored throughout the project to mitigate the effect of class imbalance.

The first strategy, `class_weight="balanced"` in the ML classifiers and `WeightedRandomSampler` in the DL pipeline, adjusts the importance assigned to each sample during training without modifying the dataset. This approach was applied as a baseline in both pipelines.

The second strategy, SMOTE, was applied to the ML pipeline following a suggestion to explore oversampling methods. SMOTE generated 2,534 synthetic NORMAL feature vectors to balance the training set. The results showed a moderate improvement in NORMAL recall, from 0.38 to 0.47 for Random Forest, confirming that oversampling on the feature space provides a measurable but limited benefit when the underlying descriptors have restricted discriminative power.

The third strategy, offline data augmentation, was explored exclusively in the DL pipeline. New NORMAL images were generated by applying geometric and photometric transformations to the existing training images and saving them to disk, balancing the dataset to 3,876 samples per class. The resulting model, referred to as v2, produced worse results than the baseline v1 across all metrics, with NORMAL recall dropping from 0.74 to 0.59 and overall accuracy falling from 88% to 83%. This outcome suggests that the synthetic images introduced a distribution shift that caused the model to overfit to the augmented training set rather than learning more generalisable representations. The v1 model, trained without offline augmentation, was therefore selected as the definitive model for this project.

4 Results

4.1 ML Pipeline Results

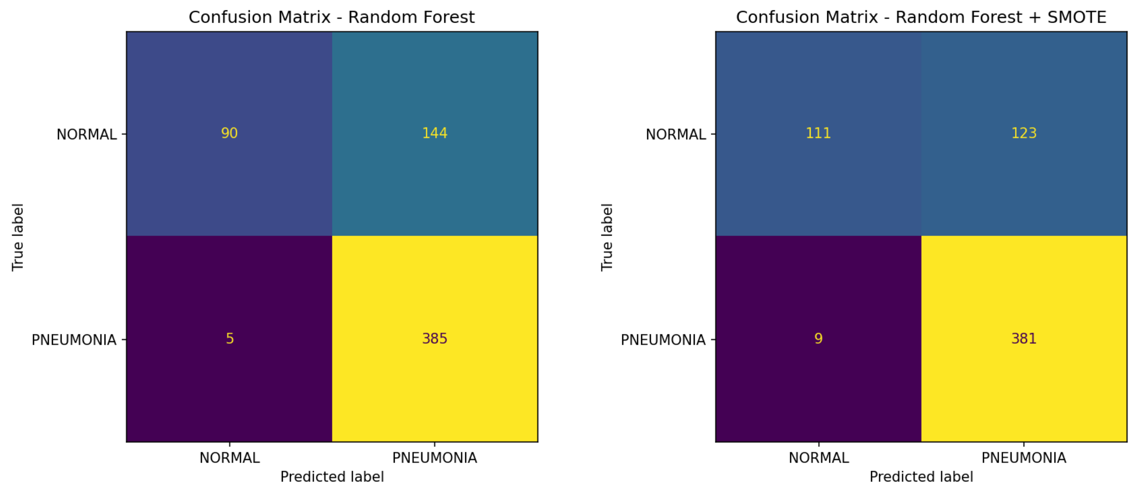
Table 2 presents the classification results for all ML models on the test set, both before and after applying SMOTE oversampling.

Table 2: ML pipeline evaluation results on the test set (624 images).

Model	Acc.	NOR. Recall	PNE. Recall	NOR. F1
Random Forest	76%	0.38	0.99	0.55
Random Forest + SMOTE	79%	0.47	0.98	0.63
XGBoost	75%	0.36	0.99	0.52
XGBoost + SMOTE	77%	0.42	0.97	0.58

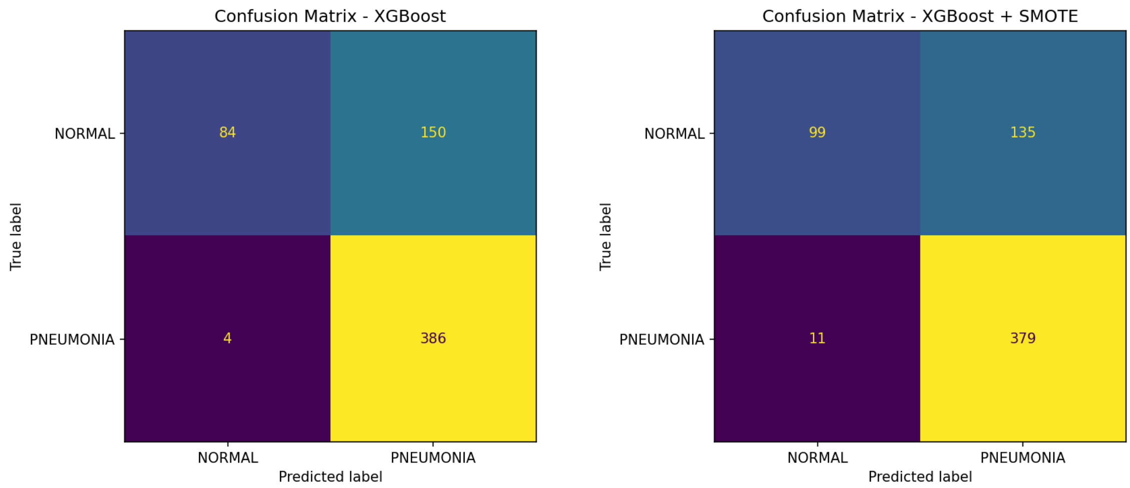
All models achieve very high PNEUMONIA recall, consistently above 0.97, reflecting the bias introduced by the class imbalance in the training set. NORMAL recall is significantly lower across all configurations, indicating that the handcrafted texture descriptors struggle to reliably distinguish healthy lung tissue from pneumonia cases. SMOTE provides a consistent improvement in NORMAL recall for both classifiers, though the gains remain modest. This suggests that the limitation lies not only in the class imbalance but also in the intrinsic discriminative capacity of the 58-dimensional feature vectors.

Figures 2 and 3 show the confusion matrices for Random Forest and XGBoost before and after SMOTE.



(a) Random Forest (b) Random Forest + SMOTE

Figure 2: Confusion matrices for Random Forest before and after SMOTE.



(a) XGBoost (b) XGBoost + SMOTE

Figure 3: Confusion matrices for XGBoost before and after SMOTE.

4.2 Feature Importance Analysis

Figure 4 shows the top 20 most important features identified by the Random Forest classifier. Contrast-based GLCM features dominate the ranking, with `Contrast_d3_a2` and `Contrast_d1_a2` occupying the first two positions with importance scores of 0.107 and 0.089 respectively. Homogeneity features also appear prominently, while correlation and energy features contribute less to the classification decision.

This finding has a direct clinical interpretation: the presence of pneumonia introduces opacities in the lung fields that create sharp intensity transitions between inflamed tissue and surrounding air. These transitions are captured by contrast-based descriptors, confirming that the GLCM contrast is the most discriminative physical texture property for pneumonia detection in this dataset. LBP features contribute to the classification but in a more distributed manner, with no single pattern dominating individually.

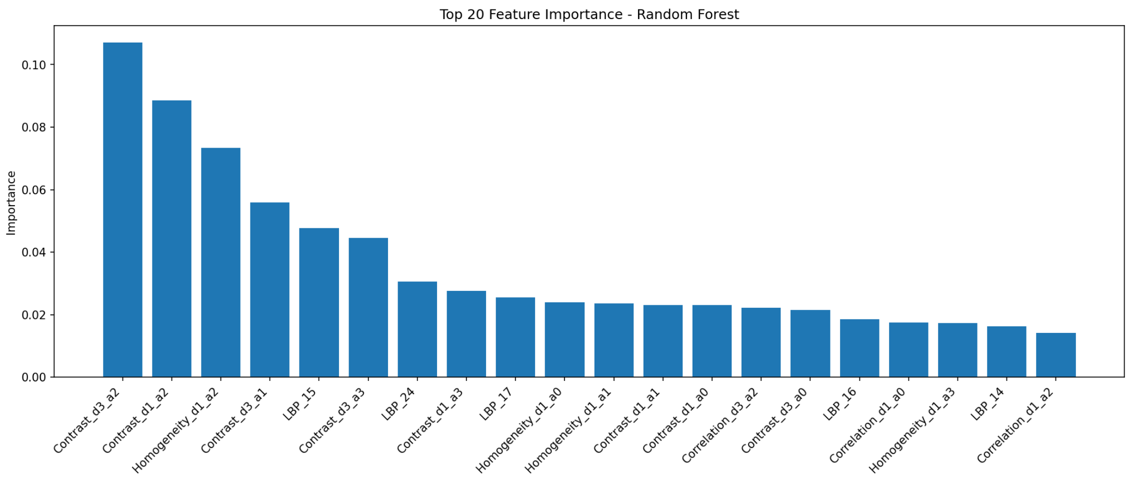


Figure 4: Top 20 most important features ranked by Random Forest. Contrast-based GLCM descriptors dominate the ranking.

4.3 DL Pipeline Results

Table 3 presents the evaluation results for DenseNet121 v1 and v2 on the test set.

Table 3: DL pipeline evaluation results on the test set (624 images).

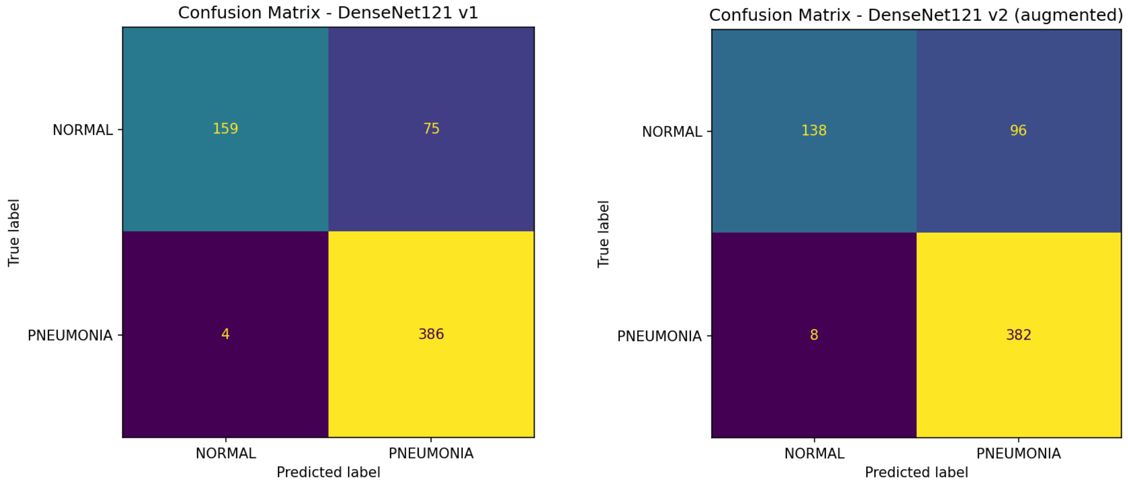
Model	**Acc.**	**NOR. Recall**	**PNE. Recall**	**NOR. F1**
DenseNet121 v1	88%	0.74	0.97	0.83
DenseNet121 v2 (augmented)	83%	0.59	0.98	0.73

DenseNet121 v1 achieves 88% accuracy with a NORMAL recall of 0.74, representing a substantial improvement over the best ML result of 79% accuracy and 0.47 NORMAL recall. The model produces 62 false positives, healthy patients classified as having pneumonia, and only 10 false negatives, pneumonia cases classified as normal. From a clinical perspective, false negatives are the more dangerous error, as they correspond to undetected disease. A PNEUMONIA recall of 0.97 means that only 1 in approximately 39 real pneumonia cases is missed by the model.

Figure 5 shows the training and validation curves across both phases. The vertical dashed line marks the transition from Phase 1 to Phase 2. Train loss decreases consistently throughout the 25 epochs, while validation accuracy remains stable between 87% and 89% during Phase 1. The brief instability visible at the phase transition is expected behaviour when additional layers are unfrozen, as the optimiser adapts to a larger set of trainable parameters.

Figure 5: Training and validation loss and accuracy across both training phases. The dashed line marks the start of fine-tuning in Phase 2.

Figure 6 shows the confusion matrices for DenseNet121 v1 and v2.



(a) DenseNet121 v1

(b) DenseNet121 v2 (augmented)

Figure 6: Confusion matrices for DenseNet121 v1 and v2.

4.4 Grad-CAM Visualization and Clinical Interpretation

Figure 7 shows the Grad-CAM heatmaps generated for representative test cases. For NORMAL cases, the model focuses primarily on the central thoracic region, where healthy lung fields show clear air-filled structures and well-defined anatomical boundaries. For PNEUMONIA cases, the activation maps concentrate on the lower lung zones, particularly where opacities and areas of increased density are present, which is consistent with the typical radiological presentation of pneumonia.

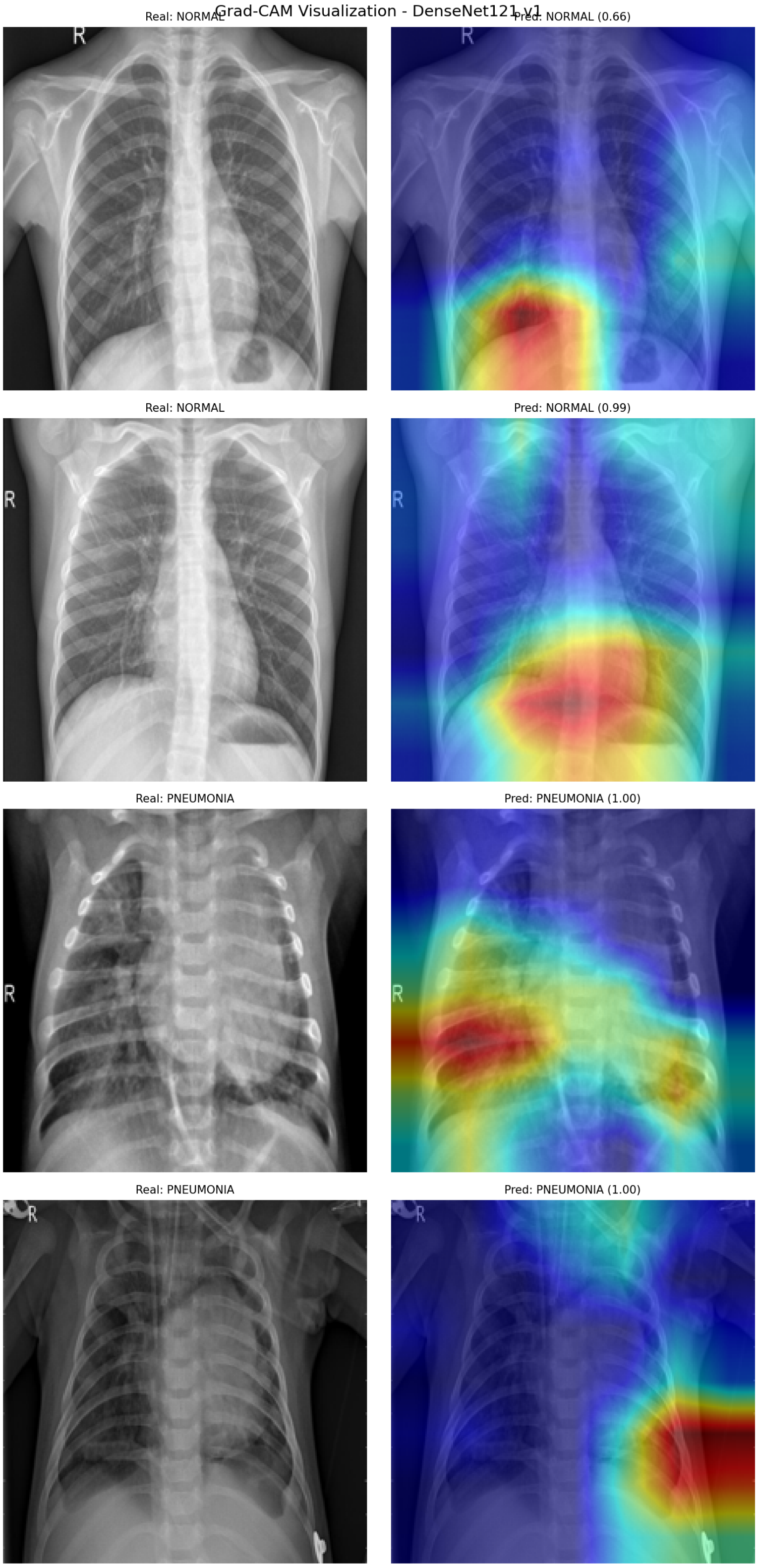


Figure 7: Grad-CAM heatmaps for representative test cases. Left column: original X-ray. Right column: Grad-CAM overlay. Rows 1-2: NORMAL cases. Rows 3-4: PNEUMONIA cases. Red regions indicate the areas most influential in the model's decision.

4.5 Comparative Analysis

Table 4 presents a unified comparison of all models evaluated in this project.

Table 4: Comparative results across all models.

Model	Acc.	NOR. Recall	PNE. Recall	NOR. F1
Random Forest	76%	0.38	0.99	0.55
Random Forest + SMOTE	79%	0.47	0.98	0.63
XGBoost	75%	0.36	0.99	0.52
XGBoost + SMOTE	77%	0.42	0.97	0.58
DenseNet121 v1	**88%**	**0.74**	0.97	**0.83**
DenseNet121 v2 (aug.)	83%	0.59	**0.98**	0.73

The results confirm that the Deep Learning pipeline consistently outperforms the Machine Learning pipeline across all metrics. The most significant gap is observed in NORMAL recall, where DenseNet121 v1 achieves 0.74 compared to 0.47 for the best ML model. This difference reflects the fundamental advantage of learned representations over handcrafted descriptors: while LBP and GLCM capture specific predefined aspects of texture, DenseNet121 learns to extract arbitrary combinations of visual features that are most discriminative for the task at hand.

The convergence between the Feature Importance analysis and the Grad-CAM visualisations provides an additional layer of validation. The ML pipeline identified contrast as the dominant discriminative feature, and the Grad-CAM heatmaps show that DenseNet121 focuses on regions of high contrast and textural heterogeneity within the lung fields. Both methodologies, despite their fundamentally different mechanisms, point to the same underlying visual signal as the primary indicator of pneumonia.

5 Conclusions

5.1 Key Findings

This project implemented and compared two independent pipelines for pneumonia detection from chest X-ray images. The results demonstrate that the Deep Learning pipeline consistently outperforms the Machine Learning pipeline across all evaluated metrics, with DenseNet121 v1 achieving 88% accuracy and a NORMAL recall of 0.74, compared to 79% accuracy and 0.47 NORMAL recall for the best ML configuration.

The Feature Importance analysis of the ML pipeline revealed that contrast-based GLCM descriptors are the most discriminative texture features for pneumonia detection, reflecting the physical reality that pneumonia introduces high-contrast opacities into the lung fields. This finding was independently corroborated by the Grad-CAM visualisations, which showed that DenseNet121 focuses on regions of high textural heterogeneity and contrast within the lung zones when classifying pneumonia cases. The convergence of both methodologies towards the same underlying visual signal provides a degree of cross-validation between the two approaches.

Regarding class imbalance, three strategies were explored. SMOTE provided a moderate improvement in the ML pipeline, increasing NORMAL recall from 0.38 to 0.47 for Random Forest. In the DL pipeline, offline data augmentation produced unexpectedly worse results, with NORMAL recall dropping from 0.74 to 0.59, likely due to the model overfitting to the artificially generated images. WeightedRandomSampler, used in the baseline DL model, proved to be the most effective strategy for this dataset.

5.2 Limitations

Several limitations should be acknowledged. The dataset contains a significant class imbalance that, despite the mitigation strategies applied, continues to influence model behaviour. The validation split provided with the dataset is extremely small, containing only 8 images per class, which required merging it with the training set and using the test set for all evaluation.

The ML pipeline, while interpretable, is fundamentally constrained by the expressiveness of the 58-dimensional feature vectors. LBP and GLCM capture specific predefined aspects of texture but cannot represent the full complexity of radiological patterns. The DL pipeline, despite its superior performance, remains partially opaque, and while Grad-CAM provides useful visual explanations, it does not constitute a formal guarantee of clinical correctness.

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